

Name – Ganesh Malaji Ghule
Roll No - 37

Assignment No 5

Jupyter Practical 5 Last Checkpoint: 2 hours ago (unsaved changes)  Logout

File Edit View Insert Cell Kernel Widgets Help Trusted Python 3 (ipykernel) 

        Code 

```
In [1]: import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.preprocessing import StandardScaler
from sklearn.metrics import confusion_matrix, ConfusionMatrixDisplay, classification_report, accuracy_score, precision_score, recall
```

```
In [2]: data = pd.read_csv('Social_Network_Ads.csv')
data.head(5)
```

```
Out[2]:
```

	User ID	Gender	Age	EstimatedSalary	Purchased
0	15624510	Male	19	19000	0
1	15810944	Male	35	20000	0
2	15668575	Female	26	43000	0
3	15603246	Female	27	57000	0
4	15804002	Male	19	76000	0

```
In [3]: data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 400 entries, 0 to 399
Data columns (total 5 columns):
#   Column          Non-Null Count  Dtype
---  ---
0    User ID         400 non-null   int64
1    Gender          400 non-null   object
2    Age             400 non-null   int64
3    EstimatedSalary 400 non-null   int64
4    Purchased       400 non-null   int64
dtypes: int64(4), object(1)
memory usage: 15.8+ KB
```

College_Practicals/DSBDAL/Practical 5.ipynb#

Jupyter Practical 5 Last Checkpoint: 2 hours ago (unsaved changes)  Logout

File Edit View Insert Cell Kernel Widgets Help Trusted Python 3 (ipykernel) 

        Code 

```
In [3]: data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 400 entries, 0 to 399
Data columns (total 5 columns):
#   Column          Non-Null Count  Dtype
---  ---
0    User ID         400 non-null   int64
1    Gender          400 non-null   object
2    Age             400 non-null   int64
3    EstimatedSalary 400 non-null   int64
4    Purchased       400 non-null   int64
dtypes: int64(4), object(1)
memory usage: 15.8+ KB
```

```
In [4]: data.describe()
```

```
Out[4]:
```

	User ID	Age	EstimatedSalary	Purchased
count	4.000000e+02	400.000000	400.000000	400.000000
mean	1.569154e+07	37.655000	69742.500000	0.357500
std	7.165832e+04	10.482877	34096.960282	0.479864
min	1.556669e+07	18.000000	15000.000000	0.000000
25%	1.562676e+07	29.750000	43000.000000	0.000000
50%	1.569434e+07	37.000000	70000.000000	0.000000
75%	1.575036e+07	46.000000	88000.000000	1.000000
max	1.581524e+07	60.000000	150000.000000	1.000000

```
In [5]: data.isnull().sum()
```

```
Out[5]: User ID          0
Gender          0
Age             0
EstimatedSalary 0
Purchased       0
```

Name – Ganesh Malaji Ghule
Roll No - 37

 jupyter Practical 5 Last Checkpoint: 2 hours ago (unsaved changes)  Logout

File Edit View Insert Cell Kernel Widgets Help Trusted Python 3 (ipykernel) 

          Code 

```
In [6]: data.shape
Out[6]: (400, 5)

In [7]: x = data.iloc[:,2:4]
        y = data.iloc[:,4]

In [8]: x_train, x_test, y_train, y_test = train_test_split(x, y, test_size=0.25, random_state=42)

In [9]: scale = StandardScaler()
        x_train = scale.fit_transform(x_train)
        x_test = scale.transform(x_test)

In [10]: lr = LogisticRegression(random_state = 0,solver = 'lbfgs')
         lr.fit(x_train,y_train)
         pred = lr.predict(x_test)

         print(x_test[:10])
         print('-'*15)
         print(pred[:10])

[[ 0.812419 -1.39920777]
 [ 2.0889839  0.52871943]
 [-0.95513241 -0.75656537]
 [ 1.0088136  0.76240757]
 [-0.85693511 -1.22394166]
 [-0.75873781 -0.23076704]
 [ 0.9106163  1.08372877]
 [-0.85693511  0.38266434]
 [ 0.2232352  0.14897619]
 [ 0.4196298 -0.14313399]]
-----
[0 1 0 1 0 0 1 0 0 0]
```

Name – Ganesh Malaji Ghule
Roll No - 37

Jupyter Practical 5 Last Checkpoint: 2 hours ago (unsaved changes) Logout

File Edit View Insert Cell Kernel Widgets Help Trusted Python 3 (ipykernel)

```
In [11]: print('Expected Output:',pred[:10])
print('-'*15)
print('Predicted Output:\n',y_test[:10])

Expected Output: [0 1 0 1 0 0 1 0 0 0]
-----
Predicted Output:
209    0
280    1
33     0
210    1
93     0
84     0
329    1
94     0
266    0
126    0
Name: Purchased, dtype: int64

In [15]: matrix = confusion_matrix(y_test,pred,labels = lr.classes_)
print(matrix)

tp, fn, fp, tn = confusion_matrix(y_test,pred,labels=[1,0]).reshape(-1)

[[61  2]
 [12 25]]

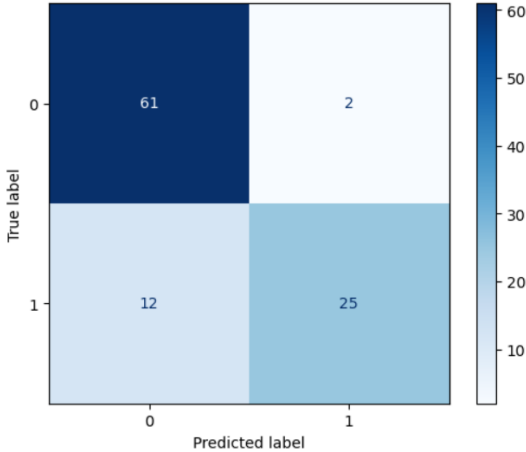
In [16]: conf_matrix = ConfusionMatrixDisplay(confusion_matrix=matrix,display_labels=lr.classes_)
conf_matrix.plot(cmap=plt.cm.Blues)
plt.show()
```



Jupyter Practical 5 Last Checkpoint: 2 hours ago (autosaved) Logout

File Edit View Insert Cell Kernel Widgets Help Trusted Python 3 (ipykernel)

```
In [16]: conf_matrix = ConfusionMatrixDisplay(confusion_matrix=matrix,display_labels=lr.classes_)
conf_matrix.plot(cmap=plt.cm.Blues)
plt.show()
```



```
In [17]: print(classification_report(y_test,pred))

precision    recall  f1-score   support
```

Name – Ganesh Malaji Ghule
Roll No - 37

